

Quick Sales Forecast with AI

I asked ChatGPT

<<Segment the customer data into groups based on purchasing behavior, engagement, and demographics. Identify high spenders (who consistently spend more), frequent buyers (customers who make regular purchases), and new customers (recently acquired customers) using the attached csv file>>

GPT-4o
Code

ChatGPT generated python code to solve the problem

```
python
Copy Code
import pandas as pd

# Load the data
data = pd.read_csv('dr_who_sales.csv')

# Convert the 'Date' column to datetime format
data['Date'] = pd.to_datetime(data['Date'])

# Grouping by 'Planet' to analyze customer behavior
customer_groups = data.groupby('Planet').agg({
    'Order Value': ['sum', 'mean', 'count'], # Total spend, average spend, and purchase frequency
    'Date': ['min', 'max'] # First and last purchase dates
}).reset_index()

# Rename columns for clarity
customer_groups.columns = ['Planet', 'Total Spend', 'Average Spend', 'Purchase Frequency', 'First Purchase', 'Last Purchase']

# Define segmentation criteria
high_spenders = customer_groups[customer_groups['Total Spend'] > customer_groups['Total Spend'].quantile(0.75)]
frequent_buyers = customer_groups[customer_groups['Purchase Frequency'] > customer_groups['Purchase Frequency'].quantile(0.75)]
new_customers = customer_groups[customer_groups['First Purchase'] >= pd.Timestamp('2243-01-01')]

high_spenders, frequent_buyers, new_customers
```

Output

output

Copy Code

```
( Planet Total Spend Average Spend Purchase Frequency First Purchase \
```

```
1 Gallifrey 735000 24500.0 30 2243-01-01
3 Krop Tor 570000 19000.0 30 2243-01-07
7 Skaro 513000 17100.0 30 2243-01-04
```

Last Purchase

```
1 2243-03-02
3 2243-03-08
7 2243-03-05 ,
```

Empty DataFrame

Columns: [Planet, Total Spend, Average Spend, Purchase Frequency, First Purchase, Last Purchase]

Index: [],

```
 Planet Total Spend Average Spend Purchase Frequency \
0      Clom    282000      9400.0          30
1    Gallifrey    735000     24500.0          30
2      Karn    330000     11000.0          30
3    Krop Tor    570000     19000.0          30
4    Midnight    150000      5000.0          30
5    New Earth    198000      6600.0          30
6 Raxacoricofallapatorius    240000      8000.0          30
7      Skaro    513000     17100.0          30
8    The Moon    429000     14300.0          30
9    Trenzalore    394500     13150.0          30
```

First Purchase Last Purchase

```
0 2243-01-28 2243-03-29
1 2243-01-01 2243-03-02
2 2243-01-19 2243-03-20
3 2243-01-07 2243-03-08
4 2243-01-22 2243-03-23
5 2243-01-25 2243-03-26
6 2243-01-16 2243-03-17
7 2243-01-04 2243-03-05
8 2243-01-10 2243-03-11
9 2243-01-13 2243-03-14 )
```

ChatGPT Results:

The segmentation of customer data based on purchasing behavior is as follows:

High Spenders

High spenders are customers whose total spend is in the top 25% of all customers. These include:

- **Gallifrey:** Total Spend = 735,000, Average Spend = 24,500, Purchase Frequency = 30
- **Krop Tor:** Total Spend = 570,000, Average Spend = 19,000, Purchase Frequency = 30
- **Skaro:** Total Spend = 513,000, Average Spend = 17,100, Purchase Frequency = 30

Frequent Buyers

No customers were identified as frequent buyers based on the top 25% of purchase frequency, as all customers have the same purchase frequency (30 purchases).

New Customers

New customers are those whose first purchase occurred on or after January 1, 2243. These include:

- **Clom:** First Purchase = 2243-01-28
- **Gallifrey:** First Purchase = 2243-01-01
- **Karn:** First Purchase = 2243-01-19
- **Krop Tor:** First Purchase = 2243-01-07
- **Midnight:** First Purchase = 2243-01-22
- **New Earth:** First Purchase = 2243-01-25
- **Raxacoricofallapatorius:** First Purchase = 2243-01-16
- **Skaro:** First Purchase = 2243-01-04
- **The Moon:** First Purchase = 2243-01-10
- **Trenzalore:** First Purchase = 2243-01-13

Let me know if you need further analysis or visualizations!

YB

I asked ChatGPT

<<Based on the characteristics of each segment. How would you tailor a marketing or sales outreach to each group base on their behaviors?>>

GPT-4o provided this output to my prompt

Here's how you can tailor marketing or sales outreach strategies for each customer segment based on their behaviors:

1. High Spenders

Characteristics: These customers consistently spend more than others, indicating a strong interest in premium products or services.

Marketing/Outreach Strategies:

- **Exclusive Offers:** Provide early access to new products, premium upgrades, or exclusive bundles tailored to their preferences.
 - **Loyalty Rewards:** Create a VIP loyalty program with perks like discounts, free shipping, or special gifts for their continued patronage.
 - **Personalized Communication:** Use personalized emails or messages highlighting their past purchases and recommending complementary or premium products.
 - **Upselling Opportunities:** Promote high-value or premium products that align with their purchasing history.
 - **Thank You Campaigns:** Send personalized thank-you notes or gifts to show appreciation for their loyalty.
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2. Frequent Buyers

Characteristics: These customers make regular purchases, indicating high engagement and consistent interest in your offerings.

Marketing/Outreach Strategies:

- **Subscription Plans:** Offer subscription-based services or products to make their purchasing process more convenient.
 - **Frequent Buyer Discounts:** Provide discounts or incentives for every nth purchase (e.g., "Buy 5, get 1 free").
 - **Engagement Campaigns:** Send regular updates about new arrivals, limited-time offers, or seasonal promotions to keep them engaged.
 - **Cross-Selling:** Recommend related or complementary products to encourage additional purchases.
 - **Feedback Requests:** Ask for their feedback on products or services to make them feel valued and involved.
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3. New Customers

Characteristics: These customers are recently acquired and may still be exploring your offerings.

Marketing/Outreach Strategies:

- **Welcome Campaigns:** Send a warm welcome email or message introducing your brand, values, and best-selling products.
- **First-Time Discounts:** Offer a special discount or freebie for their next purchase to encourage repeat business.

- **Educational Content:** Share guides, tutorials, or videos to help them understand and use your products effectively.
 - **Follow-Up Communication:** Send follow-up emails or messages after their first purchase to check their satisfaction and recommend related products.
 - **Social Proof:** Highlight customer reviews, testimonials, or case studies to build trust and encourage further engagement.
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General Recommendations

- **Data-Driven Personalization:** Use data from their purchase history, frequency, and preferences to tailor all communications.
- **Omnichannel Approach:** Reach out through multiple channels (email, SMS, social media, etc.) to ensure consistent engagement.
- **A/B Testing:** Test different offers, messages, and campaigns to see what resonates best with each segment.

By aligning your outreach strategies with the unique behaviors of each segment, you can maximize engagement, retention, and revenue. Let me know if you'd like help designing specific campaigns!